# KhataMitra

The paper udhaar notebook, replaced — a digital credit ledger for kirana shops.

BUiLDX — BuildX 36 Hours Online Hackathon · Code-A-Nova

Track: FinTech & Digital Economy

Team Tech Bro — Pintu Kumar (solo)

Live app: buildx-khatamitra.vercel.app · Code: github.com/pintu544/khatamitra

# The problem: udhaar lives on paper

How small kirana shops give credit today — in a notebook

- **No running totals.** The shopkeeper flips pages to guess who owes what — errors are common and disputes have no record to settle them.
- **No overdue view.** A 50-day-old debt looks the same as yesterday’s. Nothing flags which money is stuck the longest.
- **No systematic reminder.** Follow-up depends on memory and awkward face-to-face asks, so recovery is late — or never happens.
- **Real cost.** For a thin-margin shop, unrecovered udhaar is working capital locked up — stock that cannot be re-bought.

# The solution: KhataMitra

A focused ledger app — not another super-app

- **One khata per customer.** Running balance and a full history of every credit and payment, with dates and notes.
- **Dashboard that answers three questions.** How much is outstanding? What came back this month? Who is overdue?
- **Overdue + risk at a glance.** Rule-based repayment-risk badges sort attention to the accounts that need it first.
- **Reminder drafts, ready to copy.** Polite or firm message drafts built from the real balance — the shopkeeper sends them himself.
- **Local-first.** No sign-up, no backend, works offline; data stays in the shopkeeper’s browser.

# How it works

A day with KhataMitra

**1. Sell on credit**

Customer takes groceries now, pays later. Shopkeeper taps "+ Add credit", enters amount and a note like "atta, oil".

**2. Watch the khata**

The balance updates instantly. The dashboard total, overdue count and top debtors stay current on their own.

**3. Record payment**

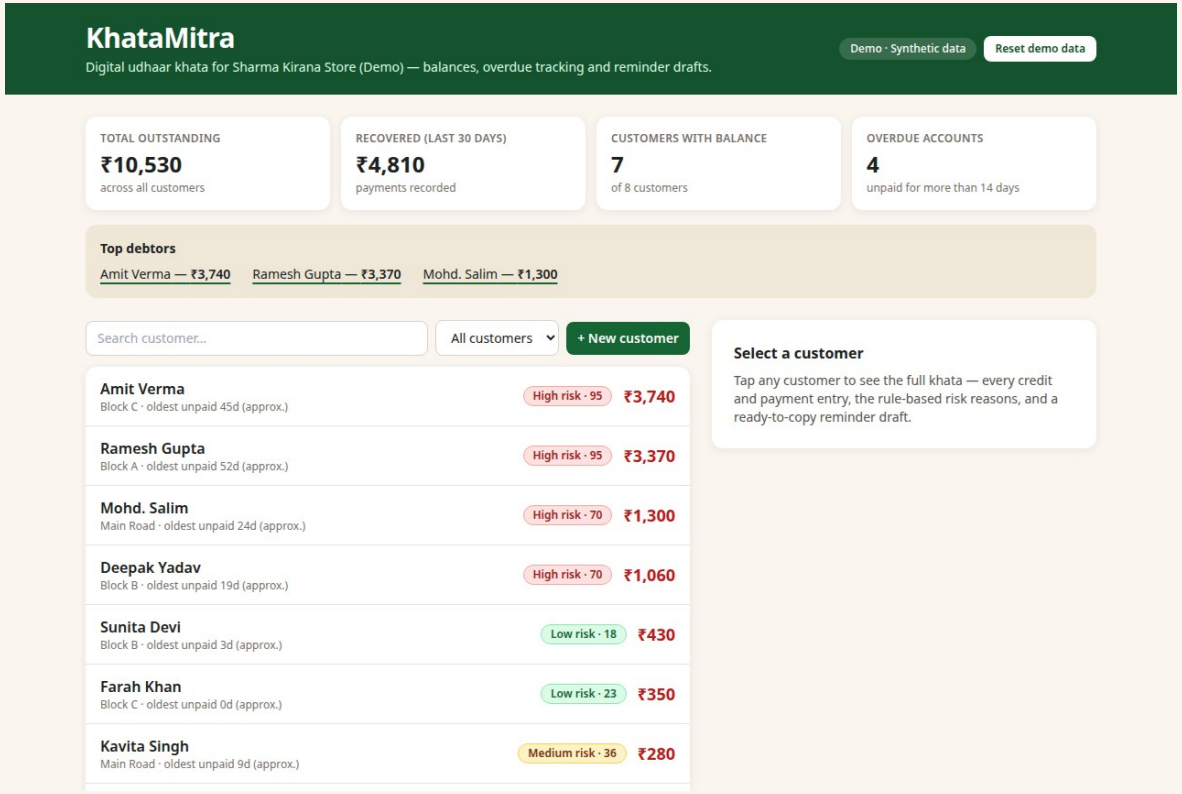
When the customer pays — cash or UPI — one tap records it against the oldest credit first.

**4. Nudge, politely**

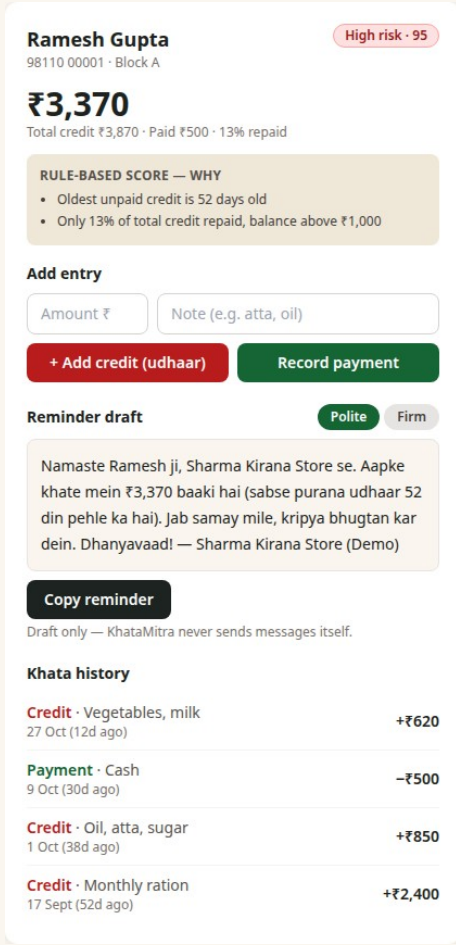
For overdue accounts, KhataMitra drafts a reminder from the actual balance and age. Copy, paste into WhatsApp, send.

# The working app

Live and deployed — screenshots from the actual build (synthetic demo data)




Dashboard — totals, top debtors, risk badges




Customer khata + reminder draft

# Key features

Everything in the live demo, nothing vapourware

- **Dashboard:** total outstanding, recovered in last 30 days, customers with balance, overdue accounts, top debtors.
- **Customer list:** search, filters (overdue only / risk band), balances sorted largest-first.
- **Add credit & record payment:** validated forms — amounts must be positive, payments cannot exceed the balance due.
- **Add customer:** name, phone, area — new khata in seconds.
- **Persistence:** localStorage — entries survive reload; one-click reset restores the demo seed.
- **Responsive:** works on the phone a shopkeeper already owns.

# Repayment-risk scoring — rule-based

No AI claims. Plain rules, shown to the user with reasons.

- **High:** oldest unpaid credit over 30 days, OR under 40% repaid with a balance above ₹1,000.
- **Medium:** oldest unpaid 15–30 days, OR 40–70% of total credit repaid.
- **Low:** recent credit and a good repayment history.
- **Always explained:** every badge in the app lists the exact reasons — the shopkeeper can judge for himself. It is a sorting aid, not a credit score and never shared with third parties.

Example from the demo: a customer 52 days overdue with 13% repaid scores High (95) — and the app says exactly why.

# Architecture & tech stack

Deliberately small — a ledger does not need a server

- **Frontend only:** React 19 + TypeScript + Vite + Tailwind CSS, deployed as a static site on Vercel.
- **State:** React context + reducer-style store, persisted to localStorage (versioned key, reset to seed).
- **Scoring engine:** one pure TypeScript function (FIFO payment allocation → oldest-unpaid age → published rules). Deterministic and testable.
- **Reminder generator:** template text filled from ledger facts only — balance and oldest-due age. Drafts only; nothing is auto-sent.
- **Why no backend:** a single shop’s khata is private, offline-first data. Sync is a roadmap item, not a demo requirement.

# Impact & roadmap

Honest scope: a working prototype with synthetic data — no user or revenue claims

- **Who it serves:** neighbourhood kirana shops that already give udhaar and lose track of it on paper.
- **What changes for them:** one clear outstanding total, overdue accounts surfaced early, and a low-awkwardness way to remind.
- **Next, if validated with real shopkeepers:** Hindi UI, UPI payment links inside reminders, printable monthly statements.
- **Then:** optional cloud backup and multi-device sync — only after a real pilot, with real feedback, before any claims are made.

# Try it, read it

Live app: https://buildx-khatamitra.vercel.app
GitHub: https://github.com/pintu544/khatamitra

The live demo opens with fictional seed data — add a credit, record a payment,
watch the risk badge move.

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Thank you.